

complemented and applied directly to the second-level NOR gate.

Example 3-10: Implement the function of Example 3-9 with NOR gates.

$$F(x, y, z) = \Sigma(0, 6)$$

$$F = x'y'z + xy'z' \text{ (SOP form)}$$

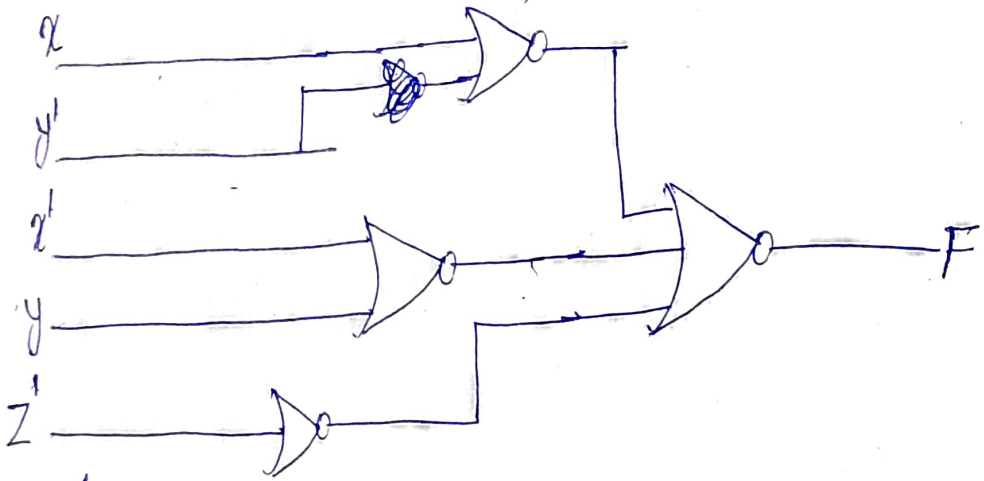
$$F' = z + x'y + xy' \text{ (SOP form)}$$

$$F = [F']' = (z + x'y + xy')' = z' \cdot (x'y)' \cdot (xy')' \text{ (De-Morgan's Theorem extended)}$$

$$= z' \cdot (x' + y') \cdot (x' + y')' \text{ (Theorem 5(b))}$$

$$= z' \cdot (x + y) \cdot (x + y)' \text{ (Theorem 3)}$$

POS form



$$F = x'y'z + xy'z' \quad F' = (x'y'z + xy'z')' = (x'y'z)' \cdot (xy'z')' \text{ (Theorem 5(a))}$$

$$= (x' + y' + z') \cdot (x + y + z') \text{ (Theorem 5(b))}$$

$$= (x + y + z') \cdot (x' + y + z) \text{ (Theorem 3)}$$

